SWISS ATMOSPHERIC CHEMISTRY POSTER DAY

PROGRAMME

9:00 - 9:45	Registration and badge collection	CAB G61
9:45 - 11:00	Welcome & Oral presentations	
11:00 - 11:15	Coffee Break	Green Floor CHN
11:15 - 12:45	Poster Session I	
12:45 - 14:00	Lunch	Food lab
14:00 - 14:45	Ice Breaker	Green Floor CHN
14:45 - 15:00	Coffee Break	
15:00 - 16:30	Poster session II	
17:00 -	Alehouse	

ORAL PRESENTATIONS

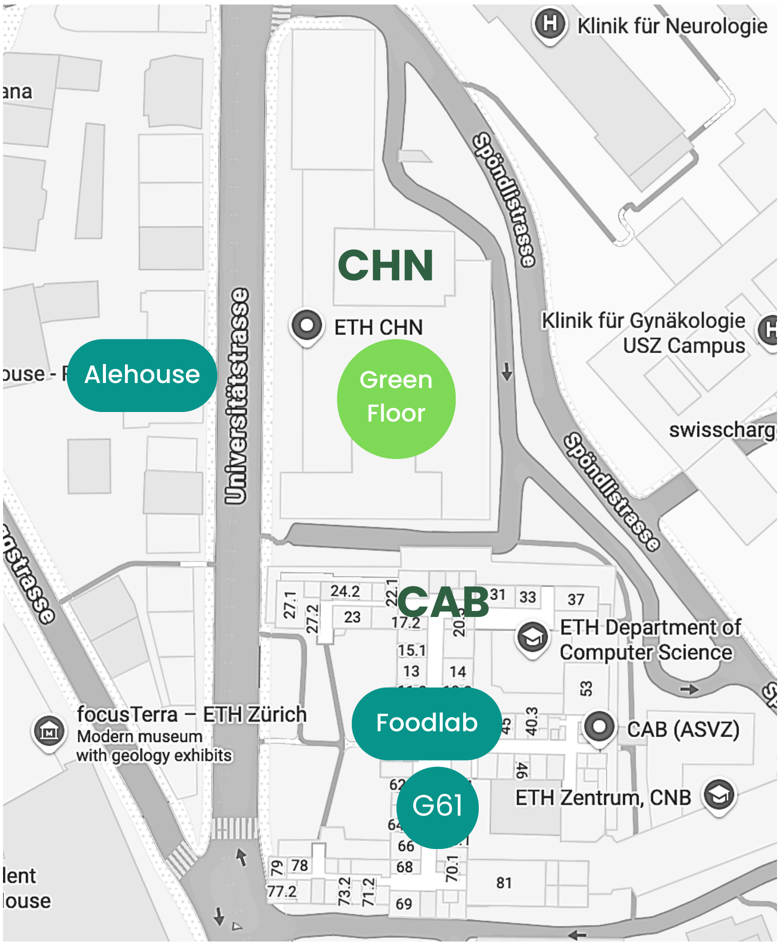
Sophie Erb: (MeteoSwiss) *SwissPollen: real-time monitoring of bioaerosols across Switzerland*

Yolanda Temel: (EPFL) *Tethered Balloon Observations of Vertical Aerosol Distributions at Neumayer III, Coastal Antarctica*

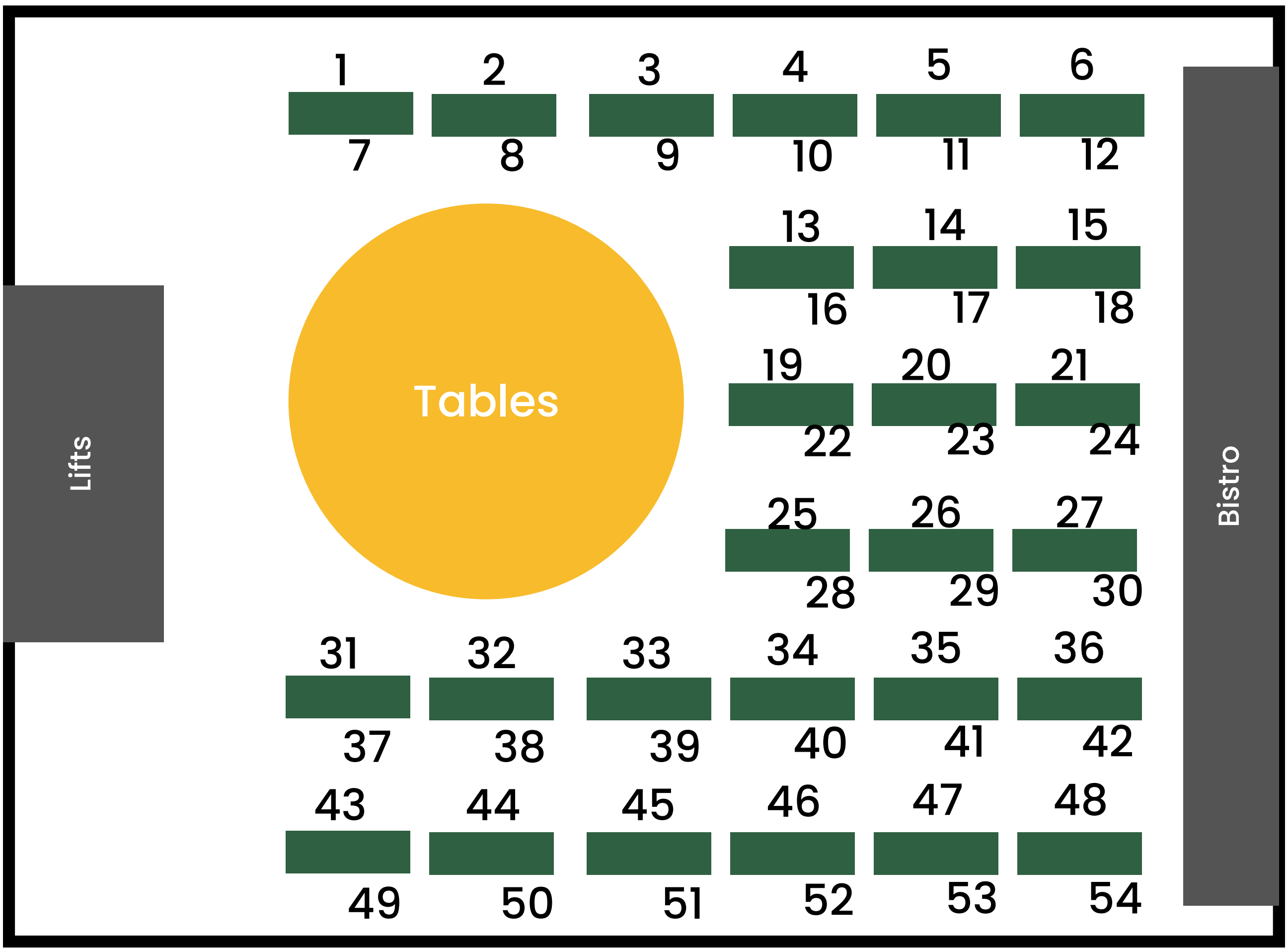
Christoph Hueglin: (Empa) *The Swiss National Air Pollution Monitoring Network (NABEL)*

Alex Valach: (Bern University of Applied Sciences) *Ammonia fluxes and deposition gradients between agricultural fields surrounding a wetland*

Atmospheric Chemistry @*ETH*



POSTERBOARD LAYOUT



Speaker: Sophie Erb

MeteoSwiss

SwissPollen: real-time monitoring of bioaerosols across Switzerland

"Pollen and fungal spore allergies affect 20–30% of Europe's population and costs between €50–150 billion per year. Traditionally, airborne pollen and spores have been monitored manually using microscopy-based counting, with results delayed by several days. Recent technological advances now enable real time monitoring by integrating artificial intelligence with advanced measurement techniques. Modern instruments can detect and classify pollen and other airborne particles—including fungal spores and aerosols larger than ~5 µm—almost instantaneously.

Since 2020, the SwissPollen network operates such devices, some SwisensPoleno. The instrument records holographic images together with fluorescence spectra and lifetimes o f single particles in real time. Deep learning algorithms classify particles by pollen type at the genus level, which is crucial information for allergic people. While early identification models relied solely on holography, recent developments show that incorporating fluorescence measurements significantly enhances classification accuracy.

Operated by MeteoSwiss, the SwissPollen network encompasses 15 stations and has delivered hourly real time pollen concentrations for seven pollen types since 2023 via its website and mobile application. These observations feed into the COSMO ART model to produce forecasts, making Switzerland the first country to provide both real time automatic pollen measurements and forecasts derived from them. The technology also shows potential for identifying additional particle types, such as Alternaria spores, iberulites associated with Saharan dust events, and possibly microplastics. As a pioneer in this field, MeteoSwiss contributes actively to European initiatives (AutoPollen, SYLVA, BioAirMet) aiming at developing and harmonising automatic pollen monitoring networks."

Speaker: Yolanda Temel

EERL (EPFL)

Tethered Balloon Observations of Vertical Aerosol Distributions at Neumayer III, Coastal Antarctica

"To better understand and parametrize clouds over the Southern Ocean and Antarctica, in-situ measurements of cloud microphysics and aerosol-cloud interactions are essential. As anthropogenic and natural continental aerosols are scarce over these regions due to their remote location, it makes them an ideal environment to study aerosol-cloud interactions.

During the ORACLES (ORigin of Antarctic CLoud particles and their Effects on the Surface radiation budget) field campaigns, in-situ vertical aerosol measurements were obtained over two summer seasons at the coastal Antarctic site of Neumayer Station III. A tethered balloon system was used to measure the particle size distributions and total concentrations in the range of 8 to 3000 nm as well as the cloud droplet size distributions up to 1000 m altitude. The thermodynamic structure of the lower atmosphere is also recorded through temperature, relative humidity as well as wind speed and wind direction data. Additionally, filter samples were also collected at different altitudes for further analysis of the aerosol chemical composition.

Here we present first results of the thermodynamic structure of the lower atmosphere and the vertical distribution of the aerosol number concentrations and size distributions at the coastal Antarctic site of Neumayer III. Measurements were conducted in different wind and weather conditions which is important because aerosol sources at Neumayer are expected to be closely linked to wind direction. Overall, the dataset including cloud-free conditions, low-level clouds and fog, and wind speeds of up to 12.5 m/s, provides valuable insight into the general summer aerosol population at Neumayer."

Speaker: Christoph Hueglin

Empa

The Swiss National Air Pollution Monitoring Network (NABEL) – bridging science and environmental policy

The Swiss National Air Pollution Monitoring Network NABEL is the nationwide air quality monitoring network operated jointly by the Swiss Federal Office for the Environment and Empa. We describe the evolution of NABEL since its founding in 1978, emphasizing its important role in monitoring air quality in Switzerland, and its contribution to international observation networks and research. The network's history, legal framework, and measurement program are presented, and exemplary time-series of air quality parameters are shown. NABEL is an excellent example for reliable, long-term air quality monitoring and the importance of such monitoring for air pollution control at both national and international levels. The NABEL network also has a long tradition of granting external users access to its infrastructure and data for their own measurement activities and research.

Speaker: Alex Valach

Bern University of Applied Sciences

Ammonia fluxes and deposition gradients between agricultural fields surrounding a wetland

"Most ammonia (NH3) emissions in Switzerland originate from agriculture and contribute to poor air quality and ecosystem degradation. Atmospheric concentrations are monitored across Switzerland using passive samplers and deposition estimated with bulk samplers. The contribution of gas phase dry deposition is calculated using ecosystem average deposition velocities. Since fluxes are in a dynamic equilibrium with surface concentrations, they produce more complex flux dynamics over time which are not captured by the static deposition velocities. National monitoring efforts are likely sufficient for large-scale annual estimates. However, since dry deposition from large sources, such as animal housings and field applications, occurs within short distances from the source, small-scale dynamics, as well as the range of net fluxes are poorly quantified.

Using new open-path NH3 sensors the SNF Sinergia ""FollowNFlow"" project aims to measure NH3 emissions from agricultural sources and net fluxes within a nearby protected wetland over 2 years to compare with neighbouring long-term passive sampler observations. Agricultural sources include animal housing and storage tanks, as well as field applications of slurry and digestates. Preliminary analyses show that agricultural emissions fall within expected ranges as modelled for the NH3 emissions inventory. Fluxes along the emission plumes show near-field deposition diminishing with distance from the source. However, net fluxes within the wetland possibly indicate bi-directional exchanges.

Results will contribute towards improving measurement methods for NH3 fluxes, as needed to assess emission reducing technologies, ecosystem protection through improved understanding of short-term horizontal gradients, as well as deposition estimation using bi-directional exchange models."

Poster Abstracts


SWISS
ATMOSPHERIC CHEMISTRY
POSTER DAY
Atmospheric Chemistry @ETH

1 Arianna Tronconi

ETH-PSI

Unmasking Cigarette Smoke in the Urban Aerosol of Barcelona, Paris and Athens

Cigarette smoke is well known to be harmful to human health, yet it remains a ubiquitous but poorly quantified source of urban aerosol. Here, the contribution of cigarette smoke to urban organic aerosol is assessed in three European cities: Barcelona, Paris, and Athens. In each city, about 180 PM10 and 180 PM2.5 filter samples were collected every second day over one year (2022–2023), yielding roughly 360 filters per city (~1000 samples in total). Filters were water-extracted and analyzed using an aerosol mass spectrometer (AMS-LToF) and an extractive electrospray ionization long-time-of-flight mass spectrometer (EESI-LToF). Positive matrix factorization (PMF) consistently retrieved a factor in all cities that contributed substantially to total OA mass and showed a chemical profile characterized by a dominant $C_{10}H_{15}N_{2}^{+}$ ion, corresponding to the protonated formula of nicotine, together with a strong temporal correlation with traffic-related markers (e.g., eBC from traffic). A subset of samples was further analyzed by EESI-Orbitrap mass spectrometry, enabling tandem mass spectrometry (MS/MS) of both the ambient $C_{10}H_{15}N_{2}^{+}$ ion and a pure nicotine standard. The resulting MS/MS spectra showed highly similar fragmentation patterns (high cosine similarity), supporting the ion identification as nicotine. In addition, EESI analysis of laboratory-generated cigarette smoke samples exhibited chemical signatures closely matching the PMF factor profiles in all three cities. These lines of evidence robustly identify a common cigarette-smoke factor across Barcelona, Paris, and Athens. The correlation with traffic markers appears to reflect human activity intensity: where people and traffic concentrate, cigarette smoke in the air does too.

2 Michael Sprenger

ETH Zurich

Where atmospheric dynamics meets atmospheric chemistry: transport modelling with Lagranto (Michael Sprenger, Heini Wernli)

"Atmospheric transport can significantly influence the chemical, microsphysical and radiative processes acting in an air parcel. For instance, an air pollutant can be released in the atmospheric boundary layer and then transported to the stratosphere where it contributes to ozone depletion; or, the air at a measurment site might be influenced by wildfires far away, i.e., affected by long-range BC transport from the source and the transformative processes along this pathway; or, Saharan dust plumes cross the Mediterranean Sea and affect visibility, but also heatwaves over Europe.

In this contribution, we provide several examples from previous and ongoing research on how atmospheric transport determines tracer distributions. The atmospheric transport is calculated with the Lagrangian Analysis Tool Lagranto based on meteorological fields from operational ECMWF and Meteoswiss products, but also from reanalysis data sets. The examples include: stratospheric ozone transport linked to stratospheric intrusions; Saharan dust events during the heatwave in July 2025; forward plume calculations from Fukushima and Cumbre Vieja volcano; wildfire BC deposition on Asian glaciers. All analysis highlight that calculation of forward and backward trajectories can substantially help understanding tracer variability and time evolution at measurement sites.

Additionally, we advertise a web-based, Lagrangian analysis tool, Lagranto.FTP, that allows kinematic forward and backward trajectories to be calculated based on ECMWF analysis (three months back in time) and forecast data (seven days forward in time). Starting locations (latitude/longitude/pressure) and starting times can arbitrarily be specified worldwide, and meteorological fields and tracers from the CAMS forecast/analysis (e.g., BC, dust, sulphate/seasalt/organic matter aerosols, CO, ozone, SO2, NOx...) can be traced along the trajectories."

3 Francesca Salteri

Paul Scherrer Institute - ETH

Investigating the role of isoprene cloud processing in SOA formation during deep convective events

"Aqueous-phase chemical processes in aerosol particles and cloud droplets can lead to the formation of secondary organic aerosol (aqSOA) and have recently been identified as a major source of SOA in the troposphere. During updrafts in deep convective clouds, volatile organic compounds (VOCs) can be taken up by cloud droplets and undergo chemical and physical transformations, with important but still poorly quantified impacts on SOA production in the upper troposphere.

Isoprene is one of the most important SOA precursors globally due to its high emission rate. The relatively high water solubility of isoprene oxidation products (iOPs) suggests that in-cloud aqSOA formation from isoprene may be significant. This hypothesis is supported by the first experimental simulation of iOP cloud processing using a wetted-wall flow reactor (WFR) by Lamkaddam et al.

To improve the understanding of in-cloud aqSOA formation from isoprene under different environmental conditions of temperature, OH exposure and NOx levels, an experimental campaign was conducted at PSI in January–February 2026. Isoprene photooxidation was performed in an oxidative flow reactor (OFR), where OH radicals production was obtained via ozone photolysis in the presence of water vapor. Isoprene oxidation's reaction products were then transferred to a WFR, where cloud conditions were simulated by a thin water microlayer.

High-resolution mass spectrometers were used to characterize organic species in the gas phase (Vocus B4 CI-ToF-MS) and in the liquid phase (EESI-ToF-MS) through online and offline measurements, respectively. We present the experimental setup, preliminary results, and a plan for further investigation.
"

4 Tuule Müürsepp

ETH Zurich

Aerosols and greenhouse gases at the dynamical tropopause: Lagrangian transport and climatology

"The upper troposphere and lower stratosphere (UTLS) serves as the transition region between the two atmospheric layers. Chemical constituents, aerosols, and water are transported in coherent airstreams or mixed from their tropospheric source regions into the UTLS. Therefore, the vertical and geographical distribution of all these constituents in the UTLS strongly depends on the transport or mixing pathways and on their (tropospheric) sources. Furthermore, during the transport constituent concentrations can be modified due to many microphysical and chemical processes. A detailed understanding about the constituent concentrations in the UTLS is needed because of their dynamic-radiative-chemical coupling, which affects atmospheric tracer distributions, the radiative budget and the dynamics of the tropopause.

We present a 10-year climatology of selected aerosol (dust, sea salt), tracer (CO) and greenhouse gas (CO2 and CH4) concentrations at the dynamical tropopause (2-pvu isosurface). We make use of the Copernicus Atmosphere Monitoring Service reanalysis datasets CAMSRA and CAMS GHG and combine these Eulerian climatologies with a Lagrangian troposphere-to-stratosphere transport (TST) climatology to determine and characterize the pathways from the constituent sources to the UTLS. This way, we analyse anomalies of aerosols and greenhouse gases at the tropopause that arise from the TST, and we compare them to seasonal atmospheric composition climatology.

We show that the TST trajectories that originated from the boundary layer (deep TST) lead to stronger anomalies at the dynamical tropopause. The concentration patterns at the dynamical tropopause for different species depend on the emissions at the surface, the exact dynamical pathway from the surface to the UTLS, and the sinks and sources along the way. For example, we demonstrate that dust concentrations at the dynamical tropopause over Asia are mostly from local dust source regions but they can be enhanced with dust advection from other regions."

5 Marco Paolini

ETH - PSI

Cloudlab: Chemical characterization of cloud composition during targeted cloud seeding

"The Beyond Cloudlab project builds on targeted cloud seeding experiments conducted within Cloudlab, which investigated the microphysical response of wintertime stratus clouds to the introduction of ice-nucleating particles (INPs). Beyond Cloudlab expands this framework by integrating detailed chemical characterization of cloud-derived samples to better constrain aerosol–cloud interactions and tracer incorporation processes. The wintertime stratus clouds at the Eriswil (BE), Switzerland, field site provide a “natural laboratory” where controlled seeding experiments can be conducted under stable atmospheric conditions.

During the field campaign, INPs such as silver iodide are introduced into clouds using a drone to induce ice formation. The resulting changes in cloud microphysics are monitored with ground-based instruments, a tethered balloon system (TBS) and a second drone.

My PhD research focuses on the chemical characterization of cloud-derived samples and background snow collected during the winter campaign. This involves developing and adapting analytical methods for low-concentration samples and performing offline analyses with techniques such as ion chromatography (IC), total organic carbon analysis (TOC), and mass spectrometry (MS/HRMS). These analyses aim to identify chemical signatures of seeding material, distinguish them from background aerosol contributions, and support the interpretation of observed microphysical changes. Moreover, initial steps are being taken toward in situ sampling of cloud water droplets and ice crystals using the tethered balloon system.

Since this research is at an early stage, the poster will outline the scientific motivation, experimental design, methodological developments, and first results."

6 Stephan Henne

Empa

Trifluoroacetate (TFA) in precipitation and surface waters in Switzerland: trends, source attribution, and budget

"Sources and budgets of the persistent, anthropogenic compound trifluoroacetate (TFA) are poorly quantified across different environmental media. Recently, the introduction of hydrofluoroolefins and the continued use of other fluorinated compounds has increased environmental levels of TFA. Here, we present concentrations of TFA observed in precipitation and surface waters in Switzerland during three years of continuous monitoring and in archived water samples, collected since 1984. Mean observed TFA concentrations ranged from 0.30 to 0.96 µgL-1 across 14 precipitation sites and from 0.33 to 0.88 µgL-1 across 9 river sites in 2021–2023 – a four-to-six-fold increase since 1996/1997. Simulated atmospheric degradation of known TFA precursors accounted for 63% (58 %–70 %) of the observed deposition (48% (41 %–54 %) hydrofluoroolefins and 15% (12 %–18 %) long-lived fluorinated gases; mean and range across sites) for sites on the Swiss Plateau. In Sitzerland, atmospheric (wet+dry) deposition of TFA amounted to 24.5 +/- 9.6Mgyr-1, whereas TFA terrestrial inputs from the degradation of plant protection products (PPP) in soils, estimated from the literature, ranged from 2.9 to 11.8Mgyr-1, depending on the assumption on degradation efficiency. TFA inputs from the degradation of PPP dominated 2–3 times over atmospheric deposition in Swiss croplands. These inputs were balanced by exports through the major rivers, 31+/-4Mgyr-1. Archived precipitation samples from the period 1986 to 2020 revealed that TFA was formed in the atmosphere before the introduction of known atmospheric precursors, whereas in the 1990s TFA deposition increased along with their concentrations. However, simulated atmospheric degradation

underestimates summertime TFA deposition five-fold. Continued use of fluorinated compounds is likely to enhance TFA deposition in the future. Additional environmental monitoring and source attribution studies are paramount for refining the assessment of TFA sources and levels for potential health and environmental risks."

7 Max Harvey

ETH Zurich

Exploring the Global Absorption from Brown Carbon in Fire Smoke

"Anthropogenic climate change is increasing both the frequency of fire-favorable weather and the likelihood of high-intensity wildfires, heightening the need to accurately represent smoke impacts on air quality, climate, and ecosystems. Yet widely used 3D models still poorly represent the evolving physicochemical and optical properties of biomass-burning (BB) organic aerosol (OA), especially the light-absorbing fraction, brown carbon (BrC), often omitting it entirely. These gaps cause systematic biases in simulated aerosol burdens and radiative effects and are responsible for large uncertainty in the net warming versus cooling effect of BB OA.

Within the GEOS-Chem (GC) global model, we develop a new BrC representation that explicitly resolves key processes from recent lab and field studies. The scheme includes long-lived dark brown carbon (d-BrC), a combustion-dependent BrC partitioning linked to black carbon-to-organic aerosol (BC: OA) emission ratios, a fire source of brown secondary organic aerosol (BrC-SOA), and a viscosity-dependent parameterization of BrC photobleaching, integrating previous individual implementations into a unified BrC module.

We assess how the revised GC representation improves agreement between modeled and observed AAOD from both AERONET and MODIS, comparing with AERONET ground-based dataand, in situ aircraft measurements (e.g., FIREX-AQ)., and MODIS satellite retrievals.

This new model scheme will enable quantification of the role of BrC from fires in present-day and future."

8 Dominik Brunner

Empa

Overview of atmospheric chemistry and transport modeling at Empa

Our group has been engaged in atmospheric chemistry and transport modeling since many years. We contributed to the development of two online coupled model systems, COSMO-ART and ICON-ART. Our main contributions to COSMO-ART were the Online Emissions Module (OEM) enabling efficient online processing of emissions, and a comprehensive cloud chemistry scheme. Our contributions to ICON-ART included the integration of OEM, detailed MOZART chemistry schemes, dry and wet deposition, and heterogeneous N2O5 chemistry. Using these model systems, we investigated the fate of secondary inorganic aerosols in Switzerland, participated in modeling intercomparison exercises, and studied future air pollution levels. We also implemented and further developed other model systems including GRAMM/GRAL to study urban air pollution and FLEXPART to study the formation and deposition of trifluoroacetic acid. This poster will present an overview of past activities and an outlook to the future of the group.

9 Rico Ka Yuen Cheung

University of Basel

Recent applications of high time-resolution online aerosol oxidative potential measurements in ambient and laboratory studies

"Aerosol oxidative potential (OP) is increasingly recognized as a more health-relevant metric than particulate mass concentration alone. An online instrument has been developed in our group for quantification of aerosol OP with temporal resolution down to 5 minutes, enabling highly time-resolved measurements in both ambient and controlled laboratory environments.

The instrument has been deployed in urban field campaigns, including roadside measurements in London and within the London Underground system, providing high time-resolution datasets under contrasting microenvironments.

In laboratory studies, the system has been applied to emissions from a range of combustion sources, including heavy-duty engines operating on low- and zero-carbon fuels. The 5-min resolution enables investigation of rapidly changing processes such as variations in engine load, combustion conditions, fuel type, and particle ageing under controlled conditions. Additional applications include brake wear particle experiments conducted in collaboration with the University of Cambridge and indoor aerosol toxicity studies performed with the University of Manchester.

These deployments demonstrate the versatility of the online OP instrument across diverse ambient and experimental settings, providing time-resolved characterization of aerosol OP for atmospheric and emission research.

10 Fabien Solmon

ETH Zurich

Integration of the Modal Aerosol Model (MAM) in the GEOS-Chem chemistry transport model: implementation, evaluation and impacts.

Aerosol process representation in global chemistry transport (CTM) and Earth system models is critical for assessing impacts on air quality, biogeochemical cycles, and climate forcing and feedbacks. While aerosol schemes describing particle size evolution, gas-phase chemistry, radiation, and cloud interactions have progressively advanced since early simplified models, their computational costs often limit their practical application for global high-resolution and climate-scale applications. The GEOS-Chem chemical transport model has traditionally employed a computationally efficient 'bulk' aerosol scheme as its standard option. Detailed sectional aerosol representations have also been developed as research options, but with substantially higher computational demands. In this context, an intermediate modal approach offers a promising niche by treating major processes governing particle size and chemical evolution at reasonable computational cost while benefiting from GEOS-Chem's advanced representation of gas-phase processes.

We present here the implementation and evaluation of the Modal Aerosol Model (MAM4, Liu et al., 2016) within the GCHP/GEOS-Chem framework to treat anthropogenic and natural aerosols. MAM is widely used in atmospheric models including, WRF-Chem, CESM, and E3SM. In our implementation, we particularly focus on the treatment of size-dependent gas-particle interactions, envisioned optionally through the MOSAIC scheme (Zaveri et al., 2008) or through a more simplified thermodynamical hybrid approach, and to the representation of secondary organic aerosols. Simulated concentrations and radiative effects of key components, are evaluated against observations for both short-term events and multi-annual simulation periods. We also compare different aerosol options in terms of accuracy and computational cost. Finally, we discuss implications of these developments for potentially facilitating community efforts toward integrating GEOS-Chem into Earth system modeling frameworks, such as CESM.

11 Yumin Li

ETH Zurich

Global Impacts of Organic Aerosol Phase State

Organic aerosol (OA) in the atmosphere can exist in liquid, semi-solid, or solid states, influenced by molecular properties and environmental conditions. However, regional and global models typically assume OA to be only in a liquid phase. Recent studies underscore that OA can present in a semi-solid or even solid state in low-temperature, dry environments. Under such conditions, increased viscosity can impede heterogeneous reaction rates by reducing diffusion. We develop a novel phase state scheme within the GEOS-Chem global model, enabling real-time simulation of OA phase states from various sources under diverse environmental conditions. Subsequently, we investigate the effects of OA phase states on heterogeneous chemical processes, including gas-particle partitioning, reactive uptake, and ice particle nucleation. Finally, our simulated OA concentrations are evaluated against global vertical profiles from aircraft observations. Our simulations indicate that on a global scale, viscosity is higher in polar regions compared to tropical regions and increases with altitude, with little to no liquid phase present above 500 hPa. Additionally, anthropogenic secondary OA (SOA) exhibits greater viscosity than biogenic SOA, hydrophobic primary OA (POA), and hydrophilic POA. The increased viscosity leads to slower gas-phase partitioning and uptake processes, thereby reducing near-source concentrations and increasing concentrations in remote areas. Considering solid-phase OA as heterogeneous ice nuclei enhances OA removal. Overall, our simulations demonstrate that incorporating phase state effect results in a reduction of OA concentrations, particularly for the more viscous SOA.

12 Zoé Le Bras

Empa

Simultaneous measurements of HCHO and (O)VOCs at an urban background site in Switzerland

13 Eric Phillip Twomey

Swiss Tropical and Public Health Institute

Capturing Space and Time: A National, High-Resolution Spatiotemporal Model of Ambient Ultrafine Particle Concentrations in Switzerland

"Objective: Despite mechanistic evidence from toxicological studies, large-scale epidemiological evidence for health effects of ultrafine particles (UFP) remains limited due to sparse monitoring. UFP exhibit large spatial and temporal contrasts, and while several spatial models have been published, few so far incorporate temporal variability. Within the framework of the MARKOPOLO ("Markers of Pollution") project, we aimed to develop a nation-wide, three-season, high-resolution (daily, 25x25m) spatiotemporal model which can and harnessed for health cohorts across Switzerland.

Methods: The project is conducting UFP measurements in Switzerland with national coverage between June 2025 and April 2026 using a mobile platform, balancing population coverage and capturing the full range of concentration contrasts. Ultrafine particle number concentration (PNC) was measured using an EPC 3787 at 1-second intervals, alongside co-pollutants (PM2.5, NO_2, black carbon), GPS position, temperature and relative humidity. Random forest models were developed to predict logged PNC using spatial and temporal environmental predictors such as altitude, meteorology, land-use, UV-radiation, traffic and road type data.

Results: During the summer measurement period (June – August 2025), 50 days and 5115km of data captured on Swiss roads yielded a median level of 7700 particles/m3 [interquartile range 5800-8500 particles/m3]. We measured within 100m of the homes of 4.5% of the Swiss population. The performance of our preliminary Swiss-wide model yielded an overall R2 of 0.90. Road length of motorways and traffic intensity and weather-related variables (temperature, humidity) were the most important predictors.

Conclusion: We provide some of the first spatiotemporal national concentration models for ultrafine PNC, allowing time-varying exposure assessment for health studies.

"

14 Luc Bessuges

ETH Zurich

Understanding selenium speciation in atmospheric deposition

Selenium (Se) is a key micronutrient for humans and animals used in the synthesis of selenoproteins which serve essential biochemical functions. Atmospheric deposition is a major source of Se for soils but the Se speciation in composition and its atmospheric pathways are still poorly understood, primarily due to the low concentrations and short lifetimes of certain species. Building on previous findings, my project aims to collect atmospheric deposition at various locations, analyze total concentrations of trace elements in wet and dry deposition and detect trace element species using various chromatographic techniques (HPLC for wet samples and TD-GC for aerosols and volatile compounds) coupled to ICP-MS/MS detection. This work will involve method development for better separation and detection of Se species. We also aim to elucidate the impact of environmental factors, such location and surroundings, seasonality and weather conditions on the atmospheric cycling of Se. Therefore we will combine chemical information with meteorological data as well as aerosol backtracking and identification of atmospheric moisture sources. The data obtained will be used as input for the global atmospheric Se model that was developed in our group. Furthermore, the methods used could also be applicable to other trace elements of interest, such as arsenic, to better understand their environmental cycling too.

15 Martin Steinbacher

Empa

Nitrogen oxide measurements serving regulatory requirements and scientific needs

"Empa and the Swiss Federal Office for the Environment have jointly operated the Swiss National Air Pollution Monitoring Network (NABEL) since its establishment in 1978. A central mandate of the network is to provide high quality, legally robust air pollution data to assess compliance with the Swiss Air Pollution Control Ordinance and to support environmental reporting and regulatory decision making. Consequently, regulatory measurements must comply with European standards such as EN 14211, which defines the reference method for nitrogen dioxide (NO_2) measurements under the Air Quality Directive (2024/2881).

EN 14211 specifies that nitrogen oxides, nitric oxide (NO) and nitrogen dioxide (NO2), shall be measured using chemiluminescence after reaction with ozone. As chemiluminescence directly detects only NO, NO2 must first be converted to NO in an upstream converter. The standard requires a conversion efficiency of at least 95 %, usually achieved with heated molybdenum converters. However, these converters are not selective for NO2 and also reduce other oxidized nitrogen species, including nitric acid, peroxyacetyl nitrates, and various organic nitrates. This may lead to systematically overestimated NO2 data, especially at rural sites where such compounds can contribute substantially to the measured signal. Despite these limitations, molybdenum converters remain the dominant technology for long-term regulatory NO2 monitoring. For scientific analysis and comparison with model output, however, such data are of limited diagnostic value.

To address this issue, several NABEL stations additionally operate selective or direct NO2 measurement techniques, including chemiluminescence with photolytic NO2 conversion, cavity attenuated phase shift spectroscopy, and direct laser absorption spectroscopy. This presentation compares NO2 concentrations measured using these techniques.

"

16 Luce Creman

IEG-USYS-ETH

Global modelling of Se speciation in atmospheric deposition

Selenium is an essential trace element for both humans, animals and vegetation. Insufficient, but also excessive Se intake is related to several health issues. For humans, crops are an important source of Se, but the availability of Se in crops depends on the concentration and chemical forms in which it is present in the soil. Se is emitted by natural and anthropogenic sources, and after atmospheric transportation and transformation it is finally deposited on the Earth's surface, including agricultural soils. Both the source and the chemical transformations affect the speciation in the deposition, and in turn determines the bioavailability of Se. In this study, we aim at producing a process-based model of the full biogeochemical cycle of Selenium, to estimate Se concentration and speciation in atmospheric deposition globally. In this poster, the preliminary research plan will be presented.

17 Michelle Jessy Müller

Empa/ETH

Estimating Urban Halocarbon Emissions in Switzerland from Atmospheric Observations

"The Montreal Protocol successfully led to the phase-out of ozone-depleting substances (ODS), thereby safeguarding the ozone layer. ODS were used as refrigerants, propellants, solvents, and in fire extinguishers. They were replaced by hydrofluorocarbons (HFCs), which are harmless to the ozone layer but are very powerful greenhouse gases. As a consequence, the Kigali Amendment to the Montreal Protocol (2016) targets a 80% reduction in HFC production and consumption by 2045. To achieve this, HFCs are being replaced with hydrofluoroolefins (HFOs) and other low-GWP alternatives.

Monitoring these substances is essential to evaluate their atmospheric impact and ensure compliance with international climate agreements. This underscores the need for reliable quantification methods. One way to estimate regional5 and global emissions halocarbon emissions is through atmospheric measurements, combined with inverse modelling. Several measurement campaigns have been conducted in Switzerland to estimate national emissions. To date, these campaigns have been limited to non-urban environments, leaving emissions within densely populated areas largely uncharacterized.

To fill this gap, different techniques were used in the city of Zurich: the tracer ratio method (TRM) and Relaxed Eddy Accumulation (REA). In the TRM, halocarbons are measured simultaneously with a co-emitted tracer (here: CO, ethylfluoride and hexafluorobutene). The ratio of analyte and tracer is then multiplied by the emission inventory (CO) or the deliberately released amount of the tracer. REA is a variation of the Eddy Covariance (EC) method that does not require a rapid-response analyzer; up- and down-draft samples are collected separately at high time resolution (ca. 10 Hz) and analyzed later. Using these approaches, the halocarbon emissions from Zurich, Switzerland's largest city, are being assessed.

18 Boxing Yang

PSI & ETH

RO2 lifetime drives biogenic new particle formation and growth in the planetary boundary layer

Biogenic new particle formation (NPF) is a major source of secondary organic aerosols (SOA) impacting both climate and human health. SOA formation involves complex atmospheric oxidation processes, making it challenging to accurately predict their yields. Current research indicates that organic oxidation initially generates peroxyl radicals (RO2), which can undergo termination via reactions with nitric oxide (NO), hydroperoxyl radical (HO2), or RO2 radicals, leading to the formation of monomer or dimer products. Depending on their volatility, these products contribute to particle nucleation and growth. However, the interactions among NO, HO2, and RO2 are governed by the HOx–ROx cycle, resulting in an intricate network of reaction pathways. This complexity makes it difficult to predict the fate of RO2 and its contribution to SOA formation. Therefore, simplifying the parameterization of RO2 fate across different termination regimes is crucial. To address this, we conducted a systematic experiment across NO:HO2:RO2 ratios relevant to the atmospheric boundary layer to investigate their influence on biogenic NPF.

19 Patrick Siegwolf

Empa

Real-time fossil CO_2 observations in urban areas by laser-based $\Delta^{14}C$-CO_2 analysis

"Urban areas are major sources of anthropogenic CO_2 emissions and play a pivotal role in emission reduction strategies. Yet, accurate quantification of local fossil emissions based solely on CO_2 mole fractions is difficult due to interfering biospheric fluxes. Radiocarbon in CO_2 (^{14}C) is a direct tracer for the share of fossil CO_2 because it is ^{14}C-depleted, unlike biogenic CO_2, which contains the current atmospheric ^{14}C level. However, urban observations at timescales sufficient for atmospheric transport models to reduce biases in emission inventories are not yet available. Currently, state-of-the art measurements are largely based on flask sampling that remain constrained in temporal and spatial coverage resulting in episodic sampling of emission patterns.

Here, we present a novel analytical system based on saturated-absorption cavity ring-down spectroscopy (ppqSense), coupled with a pre-concentration unit (NC Technologies). The system is supplemented with a quantum cascade laser-based spectrometer for CO_2 purity and $\delta^{13}C$-CO_2 analysis, required to determine $\Delta^{14}C$-CO_2. The system achieves a 30-minute temporal resolution and a repeatability of better than 10‰ in $\Delta^{14}C$-CO_2 in ambient air. We will show first time-series data of $\Delta^{14}C$-CO_2 for the Empa site in Dübendorf for ppm level quantification of fossil CO_2 contributions.

"

20 Wei Huang

Paul Scherrer Institute

Volatile Organic Compound Emissions from the Pristine Unmanaged Alaskan Boreal Forest

"Biogenic volatile organic compounds (BVOCs) are known to play a key role in secondary organic aerosol formation and subsequent cloud–climate interactions. However, current understanding which relies largely on single precursor laboratory experiments fails to capture the complex BVOC mixtures emitted by forests. In contrast to the well-studied boreal forest ecosystems in Europe, very little is known about the Alaskan boreal forest in North America. Unlike the managed boreal forest in Europe, Alaskan boreal forest is unmanaged and has numerous wildfires in the summertime. Different dominant tree species, fire dynamics, and differences in human interventions between these two types of forests may lead to distinct ecosystem-atmosphere interactions, including distinct BVOC emissions, which could lead to different climatic roles of the forest ecosystem in the atmosphere such as different strengths of aerosol formation.

During the summer of 2025, we conducted a three-month intensive field campaign at a boreal forest site in Delta Junction, Alaska. Among with other comprehensive suite of state-of-the-art instrumentation, a novel bipolar Vocus B4 Chemical Ionization-Time-of-Flight Mass Spectrometer (Vocus-B4-CI-ToF-MS, Tofwerk AG) for flux and concentrations of BVOC emissions and their oxidation products.

We will show BVOC concentrations in this pristine unmanaged boreal forest and compare with the managed ones in Europe. We will also study the effects of wildfires on BVOCs and subsequent aerosol formation and growth. The measurements at Delta Junction provide us with a valuable dataset on better understanding forest–atmosphere interactions and the climatic impacts of the unmanaged boreal forest ecosystems."

21 Sergio Cirelli

University of Bern & Oeschger Centre for Climate Change Research (OCCR)

Long-Range Atmospheric Transport of Current Used Pesticides

"The global use of synthetic pesticides, with annual production volumes exceeding 3.7 million tons, has raised significant concerns about their environmental persistence. Current-use pesticides (CUPs) have recently been detected in remote alpine and Arctic regions, challenging assumptions about their degradability and potential for long-range atmospheric transport (LRAT).

This study reconstructs the historical deposition of CUPs and legacy persistent organic pollutants (POPs) using sediment cores from two Swiss alpine lakes: Bachalpsee (2265 m a.s.l.) and Oberstockensee (1665 m a.s.l.). As these lakes are not directly impacted by pesticide application, they provide archives of pesticide deposition originating from long-range atmospheric transport.

To further assess transport potential, simulations were performed using the OECD Pov and LRTP Screening Tool across a gridded chemical partitioning space defined by Koa, Kow, and Kaw. The simulations identify distinct regions associated with differences in characteristic travel distance (CTD) estimates and show that the influence of atmospheric degradation on CTD depends strongly on a compound's position within this space. Combined with sediment records and literature field data, these results indicate that CUPs located within a specific region of this partitioning space may be more likely to undergo LRAT.

The findings highlight critical limitations in regulatory approaches that rely primarily on degradation thresholds and underscore the need for more comprehensive evaluations of pesticide transport potential in remote environments."

22 Yiwei Gong

PSI

Underestimated Short- and Long-Term Impact of Clear-Cutting on Volatile Organic Compounds in a Boreal Forest

"Boreal forests are a major source of biogenic volatile organic compounds (BVOCs), which undergo atmospheric oxidation and contribute to the formation of secondary organic aerosol (SOA) and cloud condensation nuclei. Clear-cutting, a common forest management practice involving the uniform removal of most or all trees within a designated area, can substantially alter biosphere–atmosphere interactions. In Sweden, approximately 2% of the managed forest area is harvested annually.

Here we present results from continuous observations conducted from 2020 to the present at the Norunda ACTRIS and ICOS research station in the Swedish boreal forest, where a clear-cutting event occurred in 2022 surrounding the main measurement tower. This event provided a unique opportunity to investigate the short- and long-term impacts of forest clear-cutting on atmospheric composition.

Our results show that clear-cutting significantly altered BVOC concentrations. While enhanced emissions of terpenes were expected, we also observed unexpectedly elevated concentrations of aromatic compounds, indicating that stressed boreal forests may represent an important source of aromatics. Source apportionment analysis reveals the emergence of new VOC sources during and after cutting, highlighting a more complex response of VOC emissions to forest management than previously recognized. Post-cutting factors further suggest a persistent, long-term influence on atmospheric composition. In addition, a chemical box model is used to simulate VOC oxidation processes under different clear-cutting scenarios, providing further insight into the underlying chemical mechanisms."

23 Stefan Reimann

Empa

Industrial Emissions of Ozone-Depleting Substances are Delaying the Recovery of the Stratospheric Ozone Layer

"The Montreal Protocol on Substances that Deplete the Ozone Layer has greatly restricted the global production and consumption of long-lived ozone-depleting substances (ODS). However, ODS used or consumed as feedstocks in the manufacture of other chemicals are excluded from these restrictions. This exception was granted based on the assumption that emission rates of feedstocks were very low and that production would decline in the future. Remaining emissions were expected to be too small to significantly affect the stratospheric ozone or its recovery. However, feedstock emissions are now assessed to be significantly higher than assumed and feedstock production and use has been rising.

Here, we show the full range of the usage of ODS as feedstocks in the production of hydrofluorocarbons (HFC) and hydrofluoroolefins (HFO) as well as halogenated polymers. These production schemes are then used as input to scenarios in which feedstock-related ODS emissions continue at the current elevated fraction until 2100. Without additional measures, these elevated emissions could delay the recovery of the mid-latitude stratospheric ozone layer by several years. Furthermore, limiting ODS feedstock emissions would also reduce their effect on direct radiative forcing and on climate change.

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24 Flossie Brown

ETH Zurich

Linking In-Canopy Chemistry to Above-Canopy O_3 and NO_x Fluxes in the Amazon Rainforest

25 Corina Keller

Empa

Future Ozone Exposure in Europe under Net-Zero Emission Scenario using Downscaled ICON-ART Simulations

"Tropospheric ozone is a major air pollutant that poses significant risks to human health and ecosystems. As the European Union aims to achieve net-zero greenhouse gas (GHG) emissions by 2050, it is critical to understand how emission reductions will influence near-surface ozone concentrations. Ozone formation involves complex, non-linear photochemistry of nitrogen oxides, volatile organic compounds, and meteorology, making its response to emission reductions difficult to predict.

In this study, we assess the impact of a transition to net-zero GHG emissions on near-surface ozone across Europe by comparing a reference year (2019) with a net-zero emission scenario for 2050. Simulations are performed with the ICON-ART model, configured and further developed for air quality applications. We employ the MOZART-T1 tropospheric chemistry scheme to represent key oxidation pathways. The model includes aerosol dynamics, aerosol-gas interactions, and biogenic, natural and anthropogenic emissions, providing a physically consistent representation of regional-scale atmospheric composition. However, the model's spatial resolution limits its applicability for exposure and health impact assessments.

To address this limitation, hourly surface ozone fields are downscaled to 1 km resolution using an XGBoost-based machine learning model trained on ground-based observations and a comprehensive set of chemical, meteorological, land-use, emission, and topographic predictors. This hybrid framework combines process-based atmospheric modeling with a data-driven approach to capture fine-scale spatial and temporal variability in ozone and to correct potential biases in the ICON-ART simulations. The resulting high-resolution projections are used to analyze changes in ozone distributions, extremes, and temporal variability under net-zero conditions."

26 Simone M. Pieber

WSL

Does Parental Environment Prime Offspring bVOC Responses to Heat and Drought in Scots Pine and Impact bVOC Emissions of Future Forests?

Biogenic volatile organic compounds (bVOCs) constitute a highly complex and diverse group of chemicals emitted into the atmosphere by the Earth's biosphere. Through atmospheric oxidation, they alter the mixing ratios of trace gases such as methane, carbon monoxide, and tropospheric ozone. Moreover, oxidation products contribute to aerosol formation, which plays a crucial role in Earth's radiative balance and air-quality regulation.

Projected rises in global temperatures over the coming decades are expected to produce warmer and drier conditions in Alpine regions, resulting in combined heat- and drought-stress for forest ecosystems. Understanding how trees respond to these co-occurring abiotic changes is essential for assessing impacts on atmospheric chemistry and secondary organic aerosol (SOA) properties.

We conducted controlled laboratory experiments spanning the peak growing season (July to October) to examine the effects of elevated temperature (heat, +4°C above average), reduced water availability (drought, 50% decrease in volumetric soil water content), and their combination on conifer seedlings grown from seeds collected in the Pfynwald Long-Term Irrigation Experiment (established in 2003). By using offspring from Scots pine mother trees that experienced contrasting water regimes (naturally dry versus artificially irrigated), we assessed both immediate stress responses and potential maternal-priming effects on bVOC emissions. Specifically, we investigated (i) bVOC precursors in conifer needles (secondary metabolites) and (ii) bVOC emissions in the gas phase.

To achieve this, we (i) developed an analytical method for extracting and chromatographically separating terpenes and terpenoids from conifer needles, and (ii) designed and built a novel plant chamber for bVOC gas-phase measurements - online with a PTR-ToF-MS and offline with thermodesorption-GC-MS.

We present first results illustrating terpenes and terpenoid in conifer seedlings under different maternal water availability and compare baseline conditions to heat and drought stress. We hypothesize that parental environmental history influences offspring stress responses, and that incorporating these mechanisms into Earth-system models will improve predictions of bVOC emissions and their feedbacks on atmospheric chemistry under future climate scenarios.

27 Thomas Rigny

University of Bern

Airborne profiles of per-and poly-fluoroalkyl substances (PFAS) across urban, industrial and rural landscapes using active sampling

Per- and polyfluoroalkyl substances (PFAS) are contaminants of global concern, with implications for both the environmental and human health, yet their emissions to air and indoor environments remain far less well characterized than PFAS contamination in water and soil. This project investigates how airborne PFAS mixtures and concentrations vary across sites representing contrasting source environments, including a remote alpine lake (Steinsee, 2000m elevation), an agricultural field, the University campus of Bern, a climbing gym, a wastewater treatment plant (WWTP), an industrial textile facility, and a highway corridor. Active air sampling using a Digitel high-volume aerosol sampler will provide measurements of both particle-bound and gas-phase PFAS. Preliminary hypotheses are that concentrations will be highest near WWTP and textile/fluorochemical facilities, lowest at alpine and agricultural sites, and that mixture composition will reveal source-specific fingerprints. Results will improve understanding of spatial gradients and source specific fingerprints of airborne PFAS, assess implications for local exposure and deposition, and provide evidence that can inform monitoring strategies and risk management priorities at regional scales.

28 Andrin Jörimann

PMOD/WRC & ETHZ

Separation and interplay of chemical and dynamical effects of stratospheric aerosol injection

29 Christoph Hueglin

Empa

The Swiss National Air Pollution Monitoring Network (NABEL) – bridging science and environmental policy

The Swiss National Air Pollution Monitoring Network NABEL is the nationwide air quality monitoring network operated jointly by the Swiss Federal Office for the Environment and Empa. We describe the evolution of NABEL since its founding in 1978, emphasizing its important role in monitoring air quality in Switzerland, and its contribution to international observation networks and research. The network's history, legal framework, and measurement program are presented, and exemplary time-series of air quality parameters are shown. NABEL is an excellent example for reliable, long-term air quality monitoring and the importance of such monitoring for air pollution control at both national and international levels. The NABEL network also has a long tradition of granting external users access to its infrastructure and data for their own measurement activities and research.

30 Ryan Vella

ETH Zurich

Atmospheric composition and cloud response to a weakened Atlantic Meridional Overturning Circulation

The Atlantic Meridional Overturning Circulation (AMOC) is a key regulator of Earth's climate, redistributing heat and freshwater across the Atlantic Ocean. Ongoing Arctic ice melt is expected to substantially weaken the AMOC in the future. Here, we present a global assessment of atmospheric responses to a 60% reduction in AMOC strength using a state-of-the-art chemistry–climate model. Our simulations indicate that AMOC weakening induces a pronounced slowdown of the monsoon system, leading to reduced cloud cover and suppressed deep convection over Southeast Asia. These large-scale dynamical adjustments modify surface radiation, lightning-produced NOx, and emissions of biogenic volatile organic compounds, thereby reshaping tropospheric chemistry. Concurrently, diminished precipitation substantially reduces wet deposition efficiency, allowing aerosol particles to accumulate in the atmosphere and resulting in marked increases in surface PM2.5 concentrations. We also explore the global redistribution of aerosols and the associated impacts on cloud properties.

31 Clara Nussbaumer

ETH Zurich

Carbon monoxide changes across Southern Extratropical Latitudes

32 Stuart Grange

University of Bern

Exploring the ocean and land carbon sinks with Jungfraujoch's observational records

"The global carbon budget is a critical system to understand with respect to global temperature change. An accurate carbon budget correctly allocates carbon among Earth's four key reservoirs: fossil fuel reserves, the atmosphere, the ocean, and terrestrial ecosystems. Such a budget allows for the monitoring of the health of the natural global carbon sinks. Both the ocean and land sinks have demonstrated high levels of resilience in the face of increasing anthropogenic carbon emissions and the subsequent growth of carbon dioxide (CO2) in the atmosphere. However, there are signals that these two critical carbon sinks' abilities to sequester carbon are declining.

Observations of CO2 and oxygen (O2) from the Jungfraujoch high-alpine observatory (3572 m above sea level) in the Swiss Alps will be used to constrain the carbon budget using the O2-CO2 partitioning method to illuminate the behaviour of the ocean and land sinks. High-precision measurements of CO2 and O2 are useful for partitioning because these two species are intrinsically linked through photosynthesis, respiration, and combustion processes. The O2-CO2 partitioning, along with other observational-based analyses such as the exploration of seasonal amplitudes and potentially isotopic measurements, will be used to shed light on the behaviour of the ocean and land sinks over the past decade purely from observational records, thus offering a validation and verification process for other, generally modelling-based estimates. The possible downstream climate impacts will be discussed. "

33 Qi Ran

ETH Zurich

Air Quality, Health and Economic Benefits of Transitioning Global Shipping Energy to Hydrogen

Hydrogen energy is critical for decarbonizing the hard-to-electrify shipping sector and improving air quality. However, the associated health and economic co-benefits remain unclear. Here we assess the health and economic impacts of feasible hydrogen transition scenarios in global shipping using an integrated emission-air quality-health modeling framework. Our results show that hydrogen adoption could avert approximately 130,500 (95% CI: 64,700–207,900) premature mortalities annually—75% attributable to ozone concentration reduction and 25% attributable to PM2.5 decline—with the greatest O3-attributable health gains in Southern Asia and PM2.5-attributable gains in Eastern Asia. These health benefits are estimated to save an economic cost of approximately 309 (95% CI: 148–492) billion US dollars, with India, Japan and the United States receiving the largest economic benefit. The findings highlight the potential of hydrogen-based shipping decarbonization to deliver substantial public health and economic co-benefits.

34 Iris Thurnherr

ETH Zurich

Coupling of Atmospheric Water and Trace Element Cycles in the Kara and Laptev Seas: Revealing Arctic Transport Mechanisms and Marine Biogeochemical Interactions

"The atmosphere is an important reservoir for the essential elements selenium (Se) and sulfur (S), as well as the toxic element arsenic (As). Through atmospheric deposition, these elements enter terrestrial and marine environments. Their mobility and bioavailability depend strongly on their chemical speciation. Atmospheric speciation, in turn, is influenced by the chemical forms emitted at the source and by (bio)chemical transformations during atmospheric transport.

Trace element concentrations in atmospheric deposition vary with transport pathways, source compositions, and specific weather events. More specifically, atmospheric transport of trace elements and their deposition patterns might be strongly linked to the atmospheric water cycle in particular cloud and precipitation formation, because wet deposition during precipitation is an important removal mechanism of trace elements from the atmosphere. While the importance of transport variability is well known, the underlying processes linking the water cycle with trace element transformations during emission and transport remain poorly understood.

Speciation analysis offers a powerful tool to investigate these processes because different chemical species originate from distinct emission sources and undergo different atmospheric transformations. To explore how moist processes in the atmospheric water cycle shape trace element speciation, we use the isotopic composition of water vapour as a Lagrangian process tracer.

In this study, we combine continuous in situ measurements of water vapour isotopes with speciation analyses of Se, S, and As in atmospheric aerosols collected in the Arctic marine boundary layer during the Arctic Century Expedition in the Kara and Laptev Seas (August-September 2021). Using these datasets, we assess the relative roles of local emission sources versus wet atmospheric processing in shaping observed trace element speciation patterns."

35 Anand Kumar

PSI

Photoelectron angular distribution of nitrate at aqueous solutions surfaces with varying counterions and pH conditions: a Liquid Jet XPS study

"Apparent accumulation of nitrate ions at aqueous solution surfaces has stimulated a strong interest in the field for aerosol chemistry. However, a consensus does not exist if nitrate is enhanced or depleted at the surface. In addition, another topic of debate within nitrate ion accumulation involves the degree of dissociation/ protonation of nitrate at the aqueous solution surface, which is pH dependent and remains sparsely investigated with surface specific techniques.

In this poster, we will present initial dataset collected employing Liquid jet (LJ) XPS technique in recently developed photoelectron angular distribution (PAD) mode 1. The PAD mode allows to measure elastically scattered photoelectrons detected specifically from the top layer of solutions thus circumventing the non-trivial retrieval of the depth profile based on the attenuation of photoelectrons by inelastic scattering. Here, we will present the PAD dataset collected for nitrate solutions paired with various counterions and for 1M HNO_3 and $NaNO_3$ solutions at 8 different pH between highly acidic – basic range to gain crucial information of nitrate accumulation at aqueous surfaces with chemical selectivity. This information will be complemented with surface tension data. Furthermore, for PAD-XPS at electron Kinetic energy of 50eV, a probing depth of 5-8 Å is reported. This probing depth is comparable to Non-Linear optical techniques. Thus, the data collected in this study will be further compared with previously collected sensitive information and offer clarity within the debate of nitrate accumulation.

(1) Thürmer, S.;et al., Physical review letters 2013, 111 (17), 173005.

"

36 Yolanda Temel

EERL (EPFL)

Tethered Balloon Observations of Vertical Aerosol Distributions at Neumayer III, Coastal Antarctica

"To better understand and parametrize clouds over the Southern Ocean and Antarctica, in-situ measurements of cloud microphysics and aerosol-cloud interactions are essential. As anthropogenic and natural continental aerosols are scarce over these regions due to their remote location, it makes them an ideal environment to study aerosol-cloud interactions.

During the ORACLES (ORigin of Antarctic CLoud particles and their Effects on the Surface radiation budget) field campaigns, in-situ vertical aerosol measurements were obtained over two summer seasons at the coastal Antarctic site of Neumayer Station III. A tethered balloon system was used to measure the particle size distributions and total concentrations in the range of 8 to 3000 nm as well as the cloud droplet size distributions up to 1000 m altitude. The thermodynamic structure of the lower atmosphere is also recorded through temperature, relative humidity as well as wind speed and wind direction data. Additionally, filter samples were also collected at different altitudes for further analysis of the aerosol chemical composition.

Here we present first results of the thermodynamic structure of the lower atmosphere and the vertical distribution of the aerosol number concentrations and size distributions at the coastal Antarctic site of Neumayer III. Measurements were conducted in different wind and weather conditions which is important because aerosol sources at Neumayer are expected to be closely linked to wind direction. Overall, the dataset including cloud-free conditions, low-level clouds and fog, and wind speeds of up to 12.5 m/s, provides valuable insight into the general summer aerosol population at Neumayer."

37 Célia Paolucci

PSI

HNO3 and HCl heterogeneous chemistry with CaCO3 under near-stratospheric conditions

Stratospheric Aerosol Injection (SAI) is a climate engineering scenario. Inspired by volcanic winters, the release of sulfate particles in the lower stratosphere would induce a global cooling effect. Solid particles such as calcium carbonate (CaCO3) have been proposed as they might reduce two risks of SAI: stratospheric warming and stratospheric ozone (O3) depletion. However, the heterogeneous reactions leading to O3 depletion might be more complex than first previously considered. Alkaline materials such as CaCO3 take up acidic gases. The removal of HCl, present in the stratosphere as reservoir of chlorine through the breakdown of CFCs, would be beneficial to the O3 layer while NOx removal through HNO3 uptake would be responsible for O3 depletion. We constrain some of these uncertainties by experimental work on heterogeneous chemistry of CaCO3 in presence of gaseous HCl and HNO3 under near-stratospheric conditions. Relative Humidity (RH) was set at a stratospheric value of 4% and temperature ranged between 228K and 277K. X-Ray Photoelectron Spectroscopy (XPS), a powerful surface-sensitive technique to probe solid materials within nm depths of the surface, was used. It provides quantitative elemental ratios and thus elucidates surface composition. By modelling the exponential decay of the photoelectron energy with depth, chemical composition of surface layers and hydrate states can be elucidated. T-dependence was observed, suggesting a mechanism controlled by adsorption at low T and by kinetics and diffusion at higher T. Furthermore, when exposed to both HNO3 and HCl, only HNO3 reacts with CaCO3. Such a result would suggest a more important depletion of stratospheric O3 than previously estimated.

38 Cuiqi Zhang

ETH Zurich

Direct online single-particle characterization of ice nucleatingparticles (INPs) under mixed-phase cloud (MPC) and cirrus cloud(CC) conditions at the high altitude research station Jungfraujoch(3580 m a.s.l.), Switzerland

"A scarce fraction of atmospheric aerosol particles (1 in 105 at -20 °C) can act as ice-nucleating particles (INPs) and facilitate primary ice crystal formation in clouds between -38 and 0 °C to form mixed-phase clouds, or below -38 °C to contribute to ice cloud formation. Modification of the cloud phase by INPs can lead to changes in cloud radiative properties and cloud lifetime, which further affect global precipitation patterns and energy budget. In this work, we specifically deployed online instrumentation to characterize the composition and identify the source of ambient INPs at temperatures (T) and relative humidities (RH) corresponding to mixed-phase clouds (MPCs) and ice clouds (cirrus) at JFJ from Jan. 22 to Feb. 14 in 2024. Ambient INPs were first activated into ice crystals within the Automated Horizontal Ice Nucleation Chamber (HINC-Auto) under fixed thermodynamic conditions: T = -30 °C, RHw = 104% for MPCs or T = -40 °C, RHw = 90% for cirrus. Upon exiting HINC-Auto chamber, the ice crystals were separated from liquid droplets and non-activated aerosol particles within a Pumped Counterflow Virtual Impactor (PCVI) due to their inertial difference. After sublimating the ice crystals at room temperature, these activated INPs were introduced into the Particle Analysis by Laser Mass Spectrometry-Next Generation (PALMS-NG, Jacquot et al., 2024) to determine chemical composition and size.

Our results suggest that MD is an important INP type under both mixed-phase (T = -30 ∘C, RHw = 104%) and ice cloud (T = -40 ∘C, RHw = 90%) conditions at JFJ, and Pb in biomass burning aerosol (BBA) particles might enhance their IN activity under ice cloud condition."

39 Cynthia Antossian

ETH Zurich

Viscosity, hygroscopicity, and phase state of aqueous trans-aconitic acid aerosol particles after photochemical and ozone-induced aging

Aging processes of organic aerosols, including reactions with gas phase oxidants, such as ozone (O3), as well as photochemical reactions, can significantly alter their physicochemical properties. Among these properties are viscosity, hygroscopicity, and phase state, which can affect condensed phase chemistry, gas-particle partitioning, and the ability of the aerosol particle to act as ice nucleating particles and cloud condensation nuclei, thereby affecting the Earth's energy budget and climate. While previous research has examined how photochemical aging and ozonolysis affect the physicochemical properties of organic aerosols, our study investigates the combined effect of photolysis and ozonolysis. We use aqueous trans-aconitic acid as a proxy for secondary organic aerosol particles (SOA). We observe significant mass loss in single particles levitated in an electrodynamic balance when exposed to O3 and/or UV light (375 nm), resulting from fragmentation reactions followed by the volatilization of some of the products. Viscosity measurements at 17% relative humidity revealed an increase of nearly 4 orders of magnitude after both UV exposure and combined UV and O3 exposure at 60% mass loss. Interestingly, continued UV-aging beyond 60% mass loss resulted in a viscosity decrease, whereas combined UV and O_3 exposure led to a further viscosity increase. Hygroscopicity exhibited only a modest decline after 20% mass loss during UV-aging and remained constant with further UV exposure; this reduction was less pronounced when UV-aging occurred in the presence of O_3. In addition, ozonolysis experiments performed using the microscope revealed that O3 exposure alone results in liquid-liquid phase separation.

40 Flavia Bindschedler

University of Bern

Investigating Stratospheric Intrusions at Jungfraujoch by Using Clumped CO_2 Isotopes

"Stratosphere-troposphere exchange (STE) plays a critical role in regulating the Earth's climate system by controlling the distribution of key greenhouse gases, which significantly affect radiative forcing. As climate change is expected to alter STE dynamics, improved tracers and forecasting tools are needed to constrain atmospheric budgets more precisely. This study investigates the potential of clumped CO_2 isotopes (Δ_{47}) as a tracer for stratospheric intrusions (SI) at the Jungfraujoch high-altitude research station (3571 m.a.s.l.).

The work followed three steps: forecasting SI events at Jungfraujoch, collecting air samples during intrusions, and analysing their isotopic composition. During a springtime field campaign, six flask samples were collected under varying intrusion conditions. Isotopic analysis using a low-cost isotope ratio mass spectrometer (IsoPrime100) revealed significant methodological constraints: limited CO_2 volumes allowed only single measurement cycles per flask, and systematic errors from ion source non-linearity prevented reliable Δ_{47} determination.

The results establish clear technical requirements for future work: larger sample volumes or continuous-flow measurement systems will be essential. Clumped CO_2 isotopes remain a compelling but technically demanding candidate for tracing stratosphere-troposphere exchange.

"

41 Nicole Leemann

PSI & ETH

Reduction of Nitrate through indirect Photolysis in Iron containing SOA

"Secondary organic aerosol (SOA) can mix with trace species in the atmosphere and react through multiphase or photochemistry. This aerosol aging leads to changes in composition and properties, impacting both climate and air quality. It has been shown that particulate nitrate can be reduced even above wavelengths necessary for direct photolysis. However, the detailed mechanism and the subsequent release of nitrogen oxides (NOx = NO, NO_2) and nitrous acid (HONO) to the gas phase are still poorly understood. This so-called renoxification affects the OH budget and therefore also the oxidative potential of the atmosphere. Iron can form photoactive complexes with organics that catalytically produce reactive oxygen species (ROS) through Fenton reactions, which can then interfere with particle phase redox cycles.

In Coated-Wall Flow Tube (CWFT) experiments, we expose thin films of SOA (proxy) solutions containing iron and nitrate to UVA light and detect the emitted gas phase products with several trace gas monitors and a chemical ionization mass spectrometer. We perform experiments with different coating compositions and environmental parameters, e.g. relative humidity (RH) and oxygen availability. For the SOA matrix, we mainly use citric acid as a proxy. Soon we will also do CWFT experiments comparing SOA produced with an oxidation flow reactor from different monoterpenes. The experiments will be complemented by kinetic multilayer modelling to determine reaction mechanisms and rate coefficients. First results indicate enhanced NOx and HONO yields in the presence of iron. Especially the production of NO seems to be increased, even more so at low RH."

42 Naizhong Zhang

Empa

Tracing the methane cycle by clumped isotope analysis

"As the second most important anthropogenic greenhouse gas and a major sink for stratospheric ozone, methane plays a critical role in atmospheric chemistry and climate. Methane clumped isotopic compositions (Δ13CH3D and Δ12CH2D2), defined as deviations in the abundances of two doubly substituted CH4 isotopologues from their stochastic distributions, serve as a novel proxy for CH4 formation temperature and for assessing the contribution of kinetically controlled processes . However, due to their low natural abundance, current high-resolution isotope ratio mass spectrometry (HR-IRMS) methods typically require ≥20 hours per measurement. To address these limitations, we developed a laser spectroscopic platform with a custom-built gas inlet system, enabling comparable precision within a single 20-minute run.

During an initial test phase, we optimized and validated GC-based CH4 purification and storage procedures. We now analyze CH_4 from a range of laboratory experiments and natural settings to evaluate the applicability and interpretative power of concurrent Δ13CH3D and Δ12CH2D2 analyses. We will showcase two recent examples: (1) bubble gases collected from mud volcanoes and terrestrial seeps in Italy and Romania, providing new insights into microbial overprinting of initially thermogenic hydrocarbon sources; and (2) thermogenic CH_4 trapped in source-rock pore spaces, evaluating whether the gas preserves its formation history and can be applied as a broadly applicable geo-thermometer. Our results highlight laser spectroscopy as a practical and robust tool for methane clumped isotope analysis, with strong potential to improve our understanding of constraints on source processes and ultimately, on the global methane cycle.

"

43 Thomas Krautwig

ETH Zurich

Protein Adsorption on Clay Minerals: Implications for Ice Nucleation

"Ice-nucleating particles (INPs) play a critical role in cloud microphysics by initiating ice formation in mixed-phase clouds. Ice nucleation (IN) on mineral dust, which is the most abundant atmospheric INP type, is controlled by rare surface sites that may not represent the average mineral surface. Soil dusts containing biogenic material have shown to contribute to IN at even higher temperatures than pure mineral dusts. The question therefore arises how interactions between minerals and organic macromolecules, such as proteins, modify the IN ability of either. To date, the potential of proteins to directly adsorb onto mineral surfaces and contribute to IN remain largely unknown.

Using an experimental bottom-up approach, we investigate protein adsorption on a clay mineral and its implications for immersion freezing with the Super DRoplet Ice Nuclei Counter Zurich (S-DRINCZ) offering the option of parallel cooling several well plates. The clay mineral kaolinite (0.1–0.01 wt%), which exhibits a median freezing temperature T(50) of −7.5 °C at a concentration of 0.1 wt% was mixed with the protein ferritin (0.1–0.0025 wt%), which shows a slightly higher T(50) of −6.9 °C at the same concentration in its pure form, and the mixtures were analyzed with respect to their IN ability.

Complementary UV/VIS spectroscopy is employed to determine the adsorption capacity onto kaolinite, while transmission electron microscopy (TEM) combined with EDX spectroscopy is used to localize ferritin on mineral surfaces to identify preferential adsorption sites via the iron-rich core of the protein. "

44 Paul Magyar

Empa

Adding clumped isotopes to the toolbox for understanding anthropogenic stimulation of nitrous oxide cycling

The atmospheric accumulation of nitrous oxide (N2O) is known to be driven by anthropogenic stimulation of the environmental nitrogen cycle. To ameliorate the significant effects of nitrous oxide as a greenhouse gas and driver of stratospheric ozone depletion requires a fine-scale understanding of the processes that generate and consume it, across space, time, and environmental conditions. Stable isotope measurements of δ15N, δ18O, and position-specific 15N incorporation (SP) in N2O have provided significant constraints on its sources and sinks in a variety of settings. But as our understanding of the breadth and complexity of possible microbial and environmental pathways leading to N2O rapidly evolves, existing isotopic approaches reach their limits, and additional tools that can help unravel this variability are required. We describe an approach for measuring the isotopologues 14N15N18O, 15N14N18O, and 15N15N16O readily and reproducibly using quantum cascade laser absorption spectroscopy, and we evaluate the potential utility of these N2O clumped isotopes for tracking source, sinks, and metabolic variability in N2O cycling. Nitrous oxide emissions from wastewater treatment, a significant greenhouse gas source in Switzerland, provide a prototypical application that includes the complexity of superimposed N2O cycling pathways and the challenges associated with collection and cleaning of samples from a complex aqueous system. Such measurements are a case study for future applications of N2O clumped isotopes across a broader array of natural and human-perturbed settings.

45 Thorsten Bartels-Rausch

PSI

Atmospheric Chemistry in the Cold: Molecular insights from the Lab

The resemblance of ice surfaces to supercooled water is a long-standing quest in surface science, yet the impact of this paradigm on interfacial chemistry remains poorly understood. A better understanding of interfacial hydrogen bonding is timely, not least because it is increasingly invoked to rationalize accelerated reaction rates on humid surfaces. [TB1.1]Here, we use hexylamine adsorption to directly probe chemical properties at the ice- and water- surface. Comparative X-ray photoemission measurements reveal distinct proton transfer and hydrogen bonding environments at the two interfaces.

46 Pia Karbiener

University of Basel

Airborne pathogen detection via MASC-On - Magnetic Antibody Sorting Coupled with flow Cytometry Online

"In light of the COVID-19 pandemic, airborne bioaerosol like bacteria and viruses are a major concern for human health. Therefore, monitoring of the concentrations of these bioaerosols plays a vital role in understanding disease transmission. Rapid, automated, quantitative and specific bioaerosol monitoring would allow for continuous assessment of airborne virus and bacteria at hospitals and similar locations.

In this proof-of-concept study, we develop and characterise a continuous pathogenic bioaerosol sampler and detector. The set-up consists of three main parts. In a first step (Collection phase), the bioaerosol is sampled continuously via a particle into liquid sampler. This bioaerosol liquid flow is mixed with an immunoassay (Selection phase) and then quantified in a flow cytometer (Detection phase).

We have already successfully connected all two of the three phases into one continuous, automated, and autonomous prototype for pathogenic bioaerosol monitoring. With a time resolution of 60 minutes (of which 15 are spent on sample acquisition) our instrument collects 99.5% of nebulized E. coli bacteria continuously. Flow cytometry results were confirmed with agar plate analyses. Plating can only account for colony forming units, while flow cytometry can count all labelled cells.

While other instruments for automated pathogen detection exist, these techniques lack either specificity, quantification, automation or time-resolution. This novel mix of methods attempts to combine all factors mentioned. Our next step is to optimize further and apply this method to airborne virus.

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47 Yvette Gramlich

Paul Scherrer Institute

Molecular-level insights into aerosol-fog interactions in the Italian Po Valley

The Po Valley in northern Italy is one of Europe's major winter pollution hotspots, driven by emission from industrial, agricultural and residential activities. During the cold season, stagnant air and low temperatures promote the frequent formation of dense radiation fog. These fog events modify the aerosol composition as droplets efficiently scavenge aerosol particles and trace gases, enabling aqueous phase oxidation that forms products of lower volatility. After droplet evaporation these products can remain in the particulate phase and contribute to secondary organic aerosol formation. Despite decades of research on fog in the Po Valley, the interactions between trace gases, aerosol particles and fog droplets with respect to their molecular-level chemical composition remain insufficiently understood. This knowledge gap was addressed during the Fog and Aerosol InteRAction Research Italy (FAIRARI) campaign conducted in winter and spring 2022 at an urban background site near Bologna. A comprehensive set of instruments was deployed, including a chemical ionization mass spectrometer (FIGAERO-CIMS) to characterize the chemical composition of aerosols at molecular-level. By applying Positive Matrix Factorization, we examine changes in aerosol composition during fog events to improve our understanding of how fog reshapes organic aerosol in the Po Valley.

48 Simone Brunamonti

Empa

ALBATROSS – a laser spectrometer for balloon-borne measurements of UTLS water vapor

Water vapor (H_2O) in the upper troposphere-lower stratosphere (UTLS) is of great importance to the Earth's radiative balance, yet accurate measurements of H_2O in this region (~8–25 km altitude) are still highly challenging. We developed ALBATROSS, a compact (~3.5 kg) laser spectrometer for balloon-borne measurements of UTLS H_2O. The instrument is based on a mid-IR quantum-cascade laser and a custom designed segmented-circular multipass cell with an optical path length of 6 m within a diameter of 10.8 cm. The performance of the spectrometer was assessed at UTLS-relevant conditions using SI-traceable reference gases generated by a dynamic-gravimetric permeation method. The results show that ALBATROSS achieves an accuracy better than ±1.5 % at all investigated pressures (30–250 hPa) and H_2O mixing ratios (2.5–35 ppm), and a precision better than 0.3 % at 1 s resolution. The instrument was further validated under a wide range of environmental conditions at the AIDA climate chamber during the AquaVIT-4 intercomparison. Recently, ALBATROSS was deployed in a series of atmospheric test flights in tandem with a cryogenic frospoint hygrometer (CFH). Good agreement within ±10 % was found between the two instruments up to 25 km altitude, as well as a substantially faster response time of ALBATROSS with respect to CFH. Overall, our results demonstrate the progress of ALBATROSS towards operational readiness for routine balloon deployments, and hence the potential role of mid-IR laser absorption spectroscopy in long-term monitoring of H_2O in the UTLS region.

49 Michael Sprenger

ETH Zurich

Towards an ERA5-based Lagrangian climatology of stratosphere-troposphere exchange

"The exchange of mass and tracers, e.g., ozone, across the tropopause determines the composition of the UTLS region and also dynamically couples the troposphere and stratosphere. Climatologies quantifying this cross-tropopause transport have been compiled in recent years based on reanalysis datasets, showing a strongly seasonal and regional variability. In this presentation, we compile a climatology of stratosphere-troposphere exchange (STE) based on ERA5 (1979-2023) and compare it to an ERA-Interim-based climatology. Transport from the stratopshere to the troposphere (STT) and the opposite direction (TST) will be separated.

The STE trajectories are calculated based on hourly ERA5 wind fields and the location and time where they cross the dynamical tropopause (2-PVU isosu rface) are determined. A residence time criterion is applied to distinguish transient from substantial STE, i.e., the STE trajectories have to stay a time period in the stratosphere and troposphere before and after crossing the tropopause. Furthermore, STE trajectories reaching the boundary layer (for STT) and originating from it (for TST) are separated to distinguish between vertically shallow and deep STE event. In addition to mass flux, consideration will also given to ozone

fluxes across the tropopause, and in particular to ozone fluxes reaching the planetary boundary layer.

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50 Sophie Erb

MeteoSwiss

SwissPollen: real-time monitoring of bioaerosols across Switzerland

"Pollen and fungal spore allergies affect 20–30% of Europe's population and costs between €50–150 billion per year. Traditionally, airborne pollen and spores have been monitored manually using microscopy-based counting, with results delayed by several days. Recent technological advances now enable real time monitoring by integrating artificial intelligence with advanced measurement techniques. Modern instruments can detect and classify pollen and other airborne particles—including fungal spores and aerosols larger than ~5 µm—almost instantaneously.

Since 2020, the SwissPollen network operates such devices, some SwisensPoleno. The instrument records holographic images together with fluorescence spectra and lifetimes o f single particles in real time. Deep learning algorithms classify particles by pollen type at the genus level, which is crucial information for allergic people. While early identification models relied solely on holography, recent developments show that incorporating fluorescence measurements significantly enhances classification accuracy.

Operated by MeteoSwiss, the SwissPollen network encompasses 15 stations and has delivered hourly real time pollen concentrations for seven pollen types since 2023 via its website and mobile application. These observations feed into the COSMO ART model to produce forecasts, making Switzerland the first country to provide both real time automatic pollen measurements and forecasts derived from them. The technology also shows potential for identifying additional particle types, such as Alternaria spores, iberulites associated with Saharan dust events, and possibly microplastics. As a pioneer in this field, MeteoSwiss contributes actively to European initiatives (AutoPollen, SYLVA, BioAirMet) aiming at developing and harmonising automatic pollen monitoring networks."